P. A. MANIKANDAN

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PROFESSIONAL SUMMARY

Hard-working and detail-oriented engineering student with good skills in solving problems, analyzing data, and designing systems. Good at using classroom knowledge to build real-world projects and work well in teams. Ready to use technical skills, coding knowledge, and a quick-learning attitude to contribute to an engineering team.

SKILLS

Programming Languages: Python, java,

Web Technologies: HTML5, CSS3

Database: SQL(Mysql), MongoDB
Tools & Platforms: Git, GitHub, docker, VS Code
Core Concepts: OOPs , DBMS
Soft skills: Adaptability, Team Management ,Optimistic ,Quick Leaner

EDUCATION

KonguNadu College of Engineering and Technology, Tiruchirappalli
Bachelor of Technology (B.Tech.) – Information Technology

CGPA: 7.54

Sri Vivekananda higher secondary school | 2023|
Higher Secondary (HSC)

Percentage: 88%

INTERNSHIP EXPERIENCE

Web Development Intern – Media Wave Technologies, Tiruchirappalli

- Assisted in web development and implementation of application logic using Java.
- Worked with relational databases (MySQL) and server-side workflows.
- Collaborated with team members in a real-world development environment.

PROJECTS

Sign Language to Text Converter

- **Description**: Developed a real-time sign language recognition system that converts hand gestures into readable text using Computer Vision and Machine Learning.
- Implemented gesture detection and text conversion using OpenCV and MediaPipe for accurate hand tracking.
- Designed a user-friendly interface to display recognized text in real time.
- **Technology**: python, OpenCV, Machine Learning, MediaPipe , NumPy
- **Tools:** VS Code, Git, GitHub

Smart Railway Track Obstacle Monitoring System with Intent-Based Risk Classification

- **Description**: Designed a real-time railway track monitoring system using IoT sensors and Artificial Intelligence to detect obstacles on railway tracks.
- Implemented risk classification to identify obstacle intent as accidental, careless, or intentional and generate alerts for railway authorities.
- Enhanced railway safety by providing early warning notifications to help prevent accidents and service disruptions.
- **Tools:** Arduino IDE, ESP32, Ultrasonic Sensor, IR Sensor, Git, VS Code

Certification:

- Java Programming certification
- Infosys Springboard- Frontend Development Certification
- AWS Cloud Practitioner Essentials – AWS Training & Certification (Course Completion)
- NPTEL certification - Introduction to IoT
- NPTEL certification – machine learning

HACKATHONS & PAPER PRESENTATIONS

Hackathons:

- **ST Joseph's College of Engineering –** Developed an **AI Traffic Management System** for intelligent traffic monitoring and congestion control.
- **KonguNadu College of Engineering and Technology –** Developed a **Sign Language to Text Converter** using Computer Vision and Machine Learning.
- **PSG College** — Developed **RESQ AI Emergency Response System** for rapid emergency detection and response coordination.
- **Mailam Engineering College** - Presented a **Sign Language to Text Converter** for real-time gesture recognition and text conversion.

Paper Presentations:

- **Annapoorana Engineering College –** Fake Currency Detection
  Presented an AI-based system using image processing to detect counterfeit notes by analyzing watermarks and texture.
- **Dhanalakshmi Srinivasan College –** Fake Currency Detection
  Delivered a paper on ML-based fake currency identification using feature extraction and pattern recognition techniques.